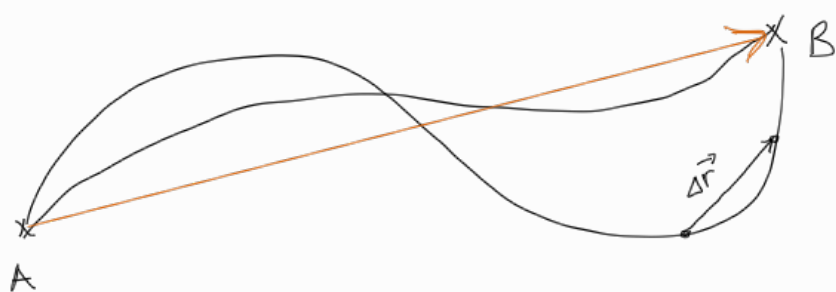
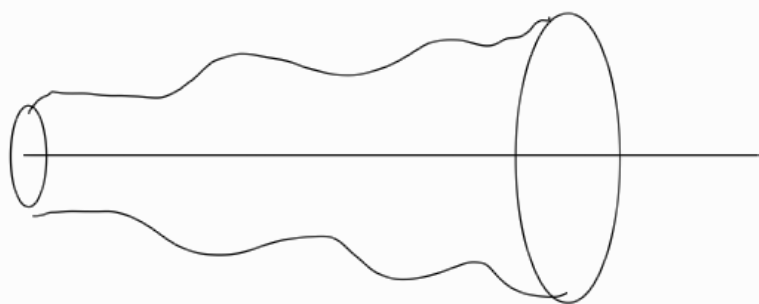


# Problema Variacional



Que camino  
optimiza la  
Distancia?

$$\mathbf{r}(t) = (x(t), y(t))$$

$$\begin{aligned} \vec{r}(0) &= A \\ \vec{r}(1) &= B \end{aligned}$$

$$\Delta t \rightarrow (\Delta x, \Delta y) \quad \text{i.e.} \quad \Delta \vec{r} = (\Delta x, \Delta y)$$

$$\Rightarrow \Delta r^2 = \Delta x^2 + \Delta y^2 \quad (\text{Instantánea línea Recta} \rightarrow \text{Pitágoras})$$

$$= \left( \left( \frac{\Delta x}{\Delta t} \right)^2 + \left( \frac{\Delta y}{\Delta t} \right)^2 \right) \Delta t^2$$

$$\Rightarrow \Delta r = \sqrt{\dot{x}^2 + \dot{y}^2} \Delta t \quad (\text{longitud de Arco})$$

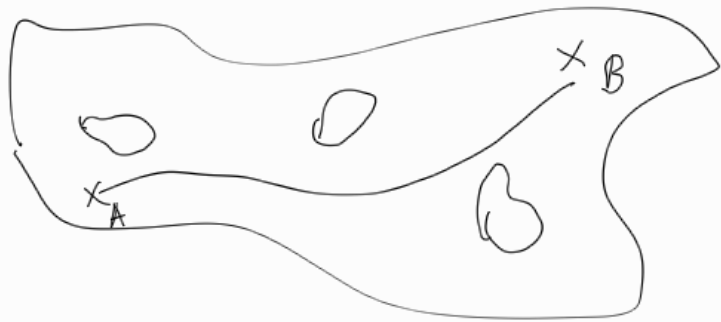
t.g. la longitud está dada por:

$$L[t \mapsto \vec{r}(t)] = \int_0^1 \sqrt{\dot{x}^2 + \dot{y}^2} dt$$

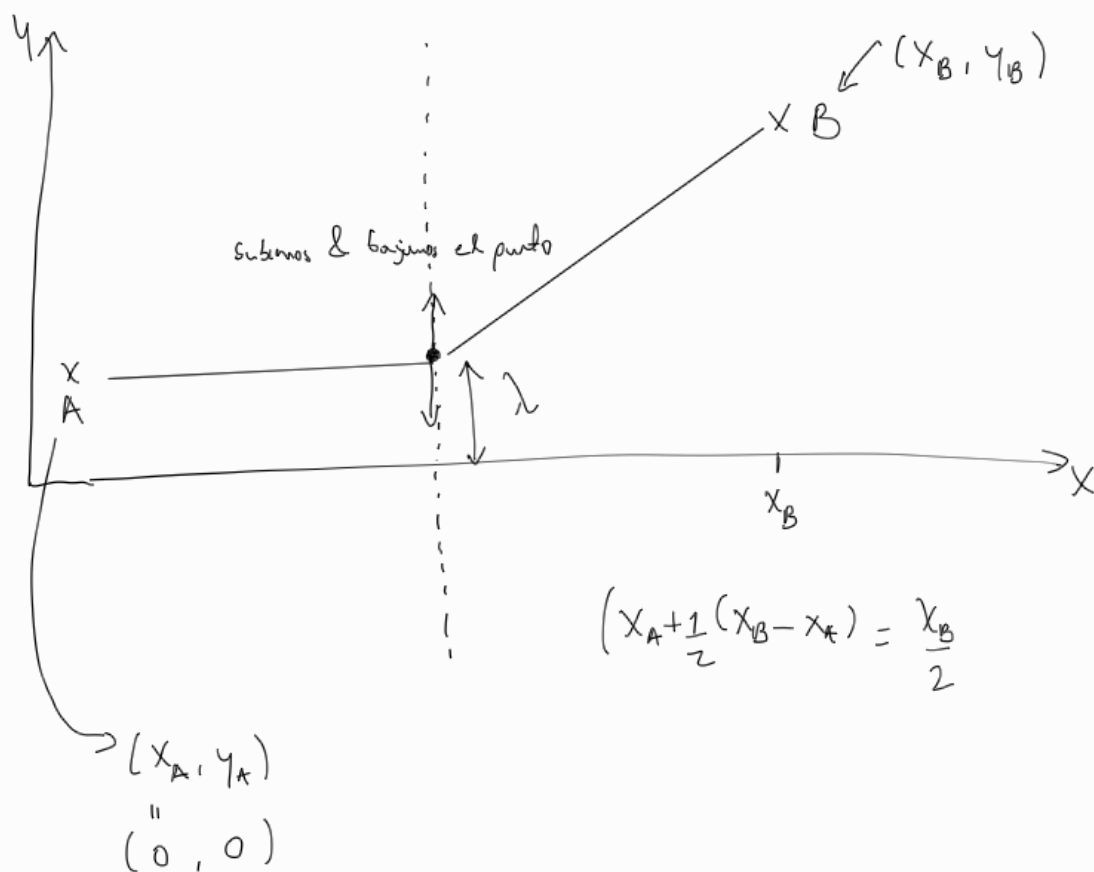
Por que es una línea Recta?



Que pasa si  
estamos en un  
espacio curvo?



~~¿~~ línea Recta...



$$\text{i.e. } (0,0) \longrightarrow \left( \frac{x_B}{2}, \lambda \right)$$

$$\left( \frac{x_B}{2}, \lambda \right) \longrightarrow (x_B, y_B)$$

$$\Rightarrow L = L_1 + L_2 = \sqrt{x_B^2 + \lambda^2} + \sqrt{\left(\frac{x_B}{2}\right)^2 + (y_B - \lambda)^2} = L(\lambda) \quad (\text{Es } L_{\text{real}})$$

$$\Rightarrow \frac{dL}{d\lambda} = 0 = \frac{2\lambda}{(x_B^2 + \lambda^2)^{1/2}} + \frac{2(\lambda - y_B)}{2\left(\left(\frac{x_B}{2}\right)^2 + (y_B - \lambda)^2\right)^{1/2}} = 0$$

$$\Rightarrow \lambda^2 \left(\frac{x_B}{2}\right)^2 + \cancel{\lambda^4} - 2\cancel{\lambda^3} y_B + \cancel{y_B^2 \lambda^2} = \lambda^2 (\lambda - y_B)^2 + (\lambda - y_B)^2 \left(\frac{x_B}{2}\right)^2$$

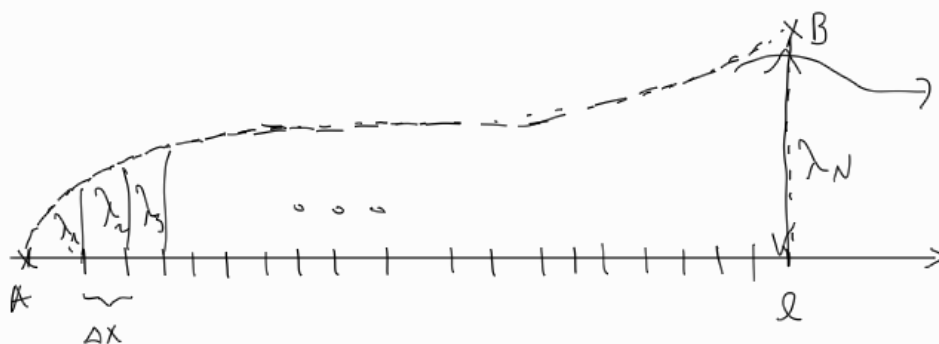
$$= \cancel{\lambda^4} - 2\cancel{\lambda^3} y_B + \cancel{y_B^2 \lambda^2} + (\lambda - y_B)^2 \left(\frac{x_B}{2}\right)^2$$

$$\Rightarrow \lambda^2 \left(\frac{x_B}{2}\right)^2 = (\lambda - y_B)^2 \left(\frac{x_B}{2}\right)^2$$

$$\Rightarrow \lambda^2 = (\lambda - y_B)^2$$

$$\Rightarrow \pm \lambda = \lambda - y_B \longrightarrow 2\lambda = y_B$$

$$\longrightarrow \underline{\lambda = \frac{y_B}{2}}$$



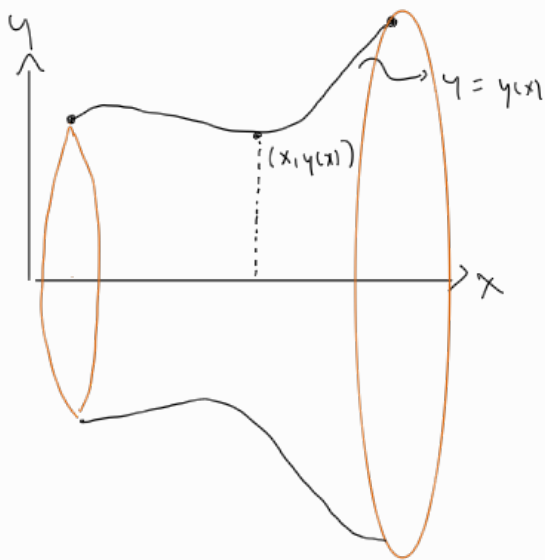
Más puntos  
 $\Rightarrow$  Más cerca de  
 la Rta.

$$\frac{l}{N} = \Delta x$$

$$\Rightarrow L(\lambda_1, \dots, \lambda_N)$$

$J$ : minimizado?

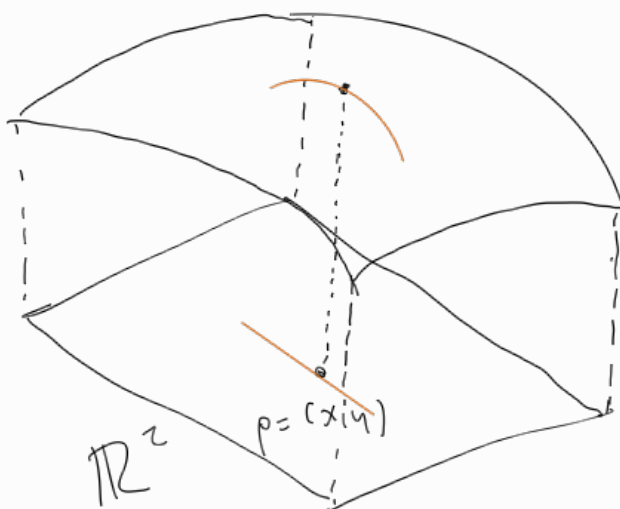
$$\frac{\partial L}{\partial \lambda_j} = 0 \quad , \quad j = 1, \dots, N$$



$$A[y] = 2\pi \int_a^b y(x) \sqrt{1 + y'(x)^2} dx$$

Que passa em Cálculo Vectorial?

$$f: \mathbb{R}^2 \longrightarrow \mathbb{R}$$

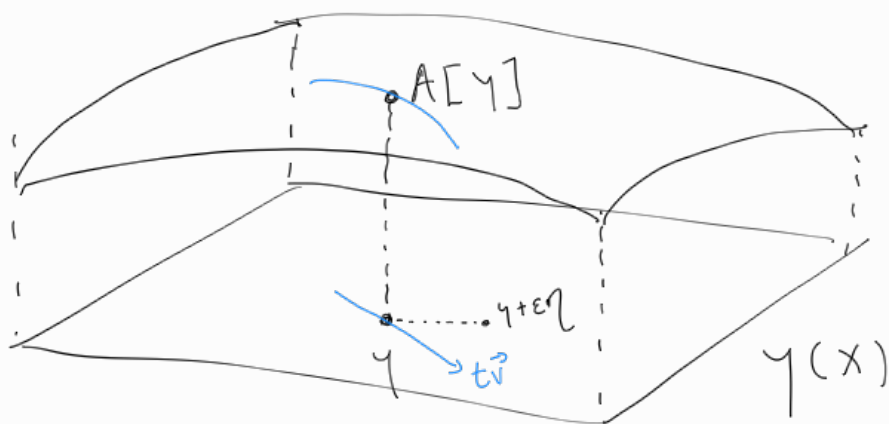


$$\text{i.e. } \vec{\nabla} f = 0$$

$$\begin{matrix} p + \Delta x \\ p + \Delta y \end{matrix} \quad \begin{matrix} \text{---} \\ \text{---} \\ \text{---} \end{matrix}$$

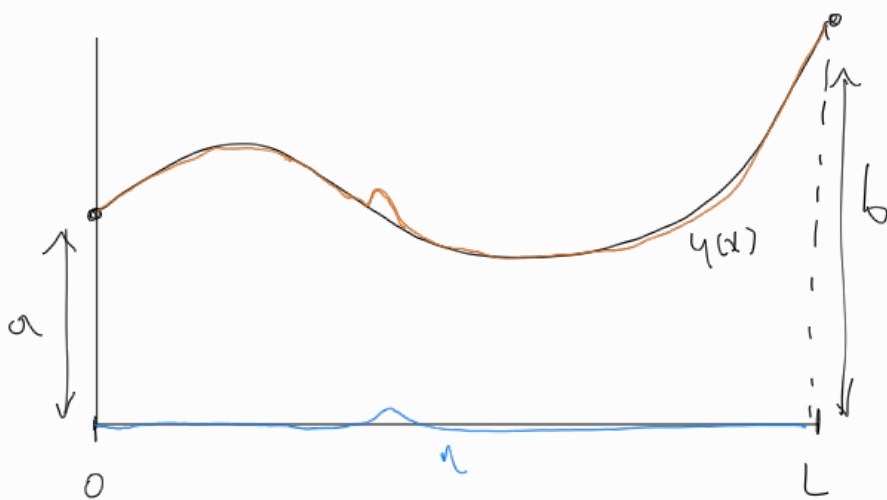
$$y = y(x)$$

"Cada punto  
Como uma função"



$$\frac{d}{dt} \bigg|_{t=0} f(p + t\vec{v}) = \vec{v} \cdot \nabla f \quad (\text{Derivada direccional})$$

Cómo me alejo de un punto?



$$\Rightarrow \begin{cases} y + \varepsilon \eta(0) = a \\ y + \varepsilon \eta(L) = b \end{cases}$$

$$\Leftrightarrow \eta(0) = 0 = \eta(L)$$

$y(x)$ : fijamos

$\eta(x)$ : Arbitraria salvo por las condiciones

$\overbrace{A[\tilde{y} + \varepsilon \eta]}^{F[\tilde{y}]}$  Suponiendo  $\tilde{y}$  la solución

$$A[y] = 2\pi \int_a^b \underbrace{y(x) \sqrt{1 + y'(x)^2}}_{f(y(x), y'(x))} dx$$

$$f(\alpha, \beta) = \alpha \sqrt{\alpha^2 + \beta^2}$$

$$\Rightarrow F[y] = \int_0^L dx f(y(x), y'(x))$$

$$\Rightarrow \frac{d}{d\varepsilon} F[\tilde{y} + \varepsilon \eta] \Big|_{\varepsilon=0} \stackrel{!}{=} 0 \quad (\text{Formulación del problema})$$

$$\Rightarrow \frac{dF}{d\lambda} \Big|_{\lambda=0} F[\tilde{y} + \lambda \eta] = 0$$

$$\text{Definimos } y_\lambda(x) = \tilde{y}(x) + \lambda \eta(x)$$

$$\Rightarrow F[y_\lambda] = h(\lambda)$$

$$\Rightarrow \frac{dh}{d\lambda} \Big|_{\lambda=0} = \frac{d}{d\lambda} \Big|_{\lambda=0} \int_0^L f(y(x), y'(x)) dx$$

$$= \int_0^L \frac{\partial f(y(x), y'(x))}{\partial \lambda} dx$$

$$= \int_0^L \left( \frac{\partial f}{\partial y} \frac{dy}{d\lambda} + \frac{\partial f}{\partial y'} \frac{dy'}{d\lambda} \right) \Big|_{\lambda=0} dx$$

pero  $\frac{dy_\lambda}{d\lambda} = \eta(x)$

$$y'_\lambda = \tilde{y}' + \lambda \eta'(x) \Rightarrow \frac{dy'_\lambda}{d\lambda} = \eta'(x)$$

$$= \int_0^L \left( \frac{\partial f}{\partial y} \eta(x) + \frac{\partial f}{\partial y'} \eta'(x) \right) dx$$



Pero  $\eta(x)$  es arbitrario, Queremos de  
encima

$$= \frac{\partial f}{\partial y} \eta(x) \Big|_0^L + \int_0^L \left( \frac{\partial f}{\partial y'} \eta(x) - \frac{d}{dx} \left( \frac{\partial f}{\partial y'} \right) \eta(x) \right) dx$$

$$= \int_a^b dx \left( \frac{\partial f}{\partial y} - \frac{d}{dx} \left( \frac{\partial f}{\partial y'} \right) \right) \eta(x) = 0$$

$$\Leftrightarrow \boxed{\frac{\partial f}{\partial y} - \frac{d}{dx} \left( \frac{\partial f}{\partial y'} \right) = 0} \quad \text{Euler-Lagrange}$$

en física:

$$\begin{aligned} x &\longrightarrow t \\ f &\longrightarrow \mathcal{L} \\ y &\longrightarrow q \\ y' &\longrightarrow \dot{q} \end{aligned}$$

$$\boxed{\frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial \dot{q}} \right) = \frac{\partial \mathcal{L}}{\partial q}}$$

Principio de Mínima Acción ( $\rightarrow$  Acción Estacionaria)

$$f(\lambda_1, \lambda_2) \quad y(x)$$

$$\hookrightarrow A[y] = \int_a^b f(y(x), y'(x)) dx$$

$$\text{t. q. } \delta A \stackrel{!}{=} 0$$



Supongamos

$$\gamma_\lambda = \tilde{\gamma} + \lambda \eta$$

Definiendo  $h(\lambda) := A[\gamma_\lambda] \rightarrow \frac{dh}{d\lambda} = 0$

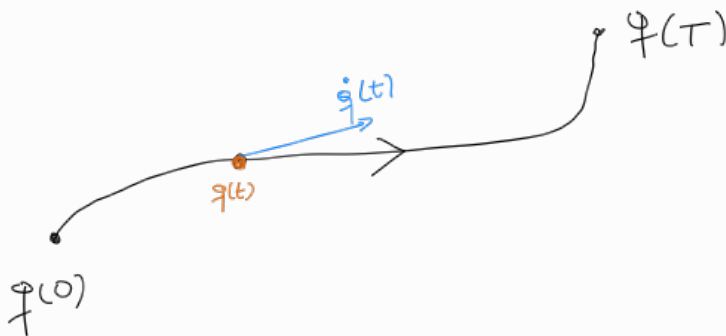
$$\frac{\partial f}{\partial y} = \frac{d}{dx} \left( \frac{\partial f}{\partial y'} \right)$$

Ahora en el contexto físico ...

• Variables Generalizadas:  $(q_1, q_2, \dots, q_N) = q$

entonces las trayectorias físicas son curvas en el espacio físico de  $q$

$$t \mapsto q(t)$$



$$\frac{dq}{dt} \equiv \dot{q}(t) = (\dot{q}_1(t), \dot{q}_2(t), \dots, \dot{q}_N(t)) \leadsto \text{Vector velocidad}$$

- $L: \mathbb{R}^N \times \mathbb{R}^N \longrightarrow \mathbb{R}$

~~$q, \dot{q}$~~

$$(x, y) \longmapsto L(x, y)$$

$$x = (x_1, \dots, x_N); \quad y = (y_1, \dots, y_N)$$

tomamos cualquier trayectoria

$$t \longmapsto q(t) = (q_1(t), \dots, q_N(t))$$

$\downarrow$

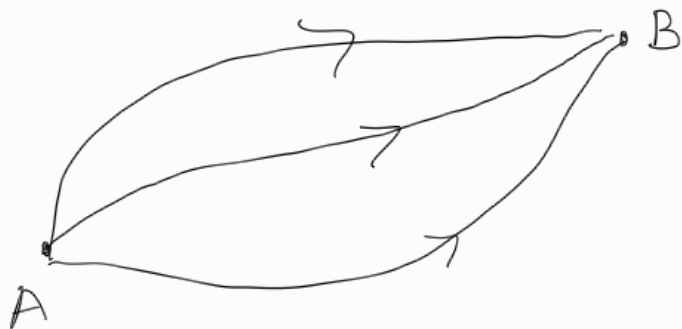
$$\dot{q}(t) = (\dot{q}_1(t), \dots, \dot{q}_N(t))$$

$\Rightarrow$

$$L(q(t), \dot{q}(t)) \in \mathbb{R}$$

Cuando trabajamos con lagrangianos sabemos que dependerá de velocidades & momentos

Vemos el camino físico:



tomamos cualquier camino

$$S[q(t)] = \int_{t_A}^{t_B} L(q(t), \dot{q}(t)) dt$$

Es lo que llamamos Acción como función del camino  $q(t)$

i.e.  $f \longrightarrow L$

$$y \longrightarrow q$$

$$y' \longrightarrow \dot{q}$$

$$x \longrightarrow t$$

Matemática

Física

$\longrightarrow$  Se vuelve un problema dinámico

$$\Rightarrow \delta S \stackrel{!}{=} 0 \quad \longleftrightarrow \quad \text{Euler-Lagrange} \quad \frac{\partial L}{\partial q_i} = \frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}_i} \right)$$

(\*)

Curiosamente hay una forma de escribir la Mecánica Cuántica con Integral de Caminos de Feynman

• Partícula libre en  $\mathbb{R}^3$

$$L(q, \dot{q}) = \frac{1}{2} m \dot{q}^2, \quad q = (q_1, q_2, q_3)$$

$$\hookrightarrow L = T - V \longrightarrow \underline{\text{Por qué?}}$$

$$\Rightarrow \frac{\partial L}{\partial q_i} = 0, \quad \frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}_i} \right) = \frac{d}{dt} (m \dot{q}_i) = m \ddot{q}_i$$

$$\Rightarrow m\ddot{q}_i = 0 \rightarrow \text{Mov. con } v = \text{cte}$$

En un caso más general

(\*) Partícula en el espacio sujeta a la Acción de un potencial  $U$

$$\Rightarrow \vec{F} = -\vec{\nabla}U \quad (\text{Newton})$$

$$\& \text{ 2da Ley de Newton: } \vec{F} = m\vec{a}$$

$$\Rightarrow m\ddot{\vec{q}} = -\vec{\nabla}U$$

$$2. \quad m\ddot{q}_i = -\frac{\partial U}{\partial q_i} \quad \left\{ \begin{array}{l} \text{Problemas Amplios de Mecánica orbital} \end{array} \right.$$

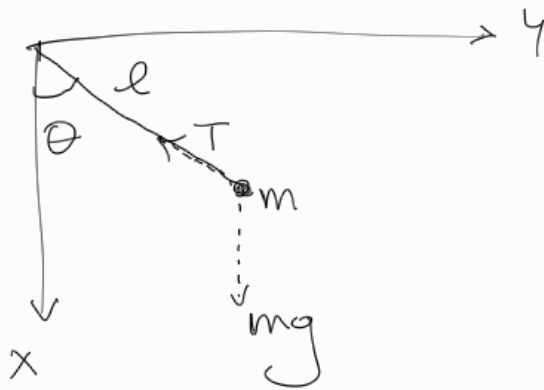
↙ sistema autónomo No depende del tiempo

$$L(q, \dot{q}) = \frac{1}{2} m \dot{\vec{q}}^2 - U(\vec{q})$$

$$\left( \begin{array}{l} \frac{\partial L}{\partial q_i} = -\frac{\partial U}{\partial q_i} \quad ; \quad \frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}_i} \right) = \frac{d}{dt} (m\dot{q}_i) = m\ddot{q}_i \end{array} \right.$$

$$m\ddot{q}_i = -\frac{\partial U}{\partial q_i} \quad (\text{Recuperamos Newton})!$$

Veamos un sistema armónico:



Newton :

$$U = -mgx$$

$$\vec{r} = (x, y)$$

→ Lagrange

$$\mathcal{L}(x, y, \dot{x}, \dot{y}) = T - U$$

$$= \frac{1}{2} m(\dot{x}^2 + \dot{y}^2) + mgx$$

El Lagrangiano No tiene ni idea  
de que son fuerzas o tensiones Ni direcciones!

①

$$\frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \frac{\partial \mathcal{L}}{\partial x} \longrightarrow m\ddot{x} = mg \longrightarrow \parallel \ddot{x} = g \left\{ \begin{array}{l} \text{Caída libre} \end{array} \right.$$

$$\textcircled{2} \frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial \dot{y}} \right) = \frac{\partial \mathcal{L}}{\partial y} \longrightarrow m\ddot{y} = 0 \longrightarrow \perp \ddot{y} = 0 \left\{ \right.$$

Ligadura :  $x^2 + y^2 = l^2$

Introducimos  $f(x,y) = x^2 + y^2 - l^2 \stackrel{!}{=} 0$  ←

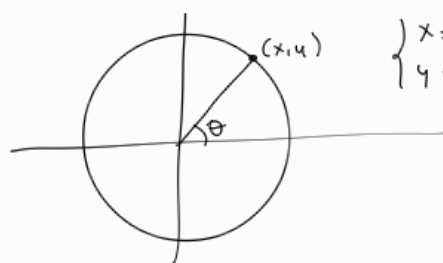
Pasamos de un círculo a una Superficie de Nivel.

⇒ Lagrangiano con Restricción

$$\mathcal{L}_r = \mathcal{L} - \lambda f = \mathcal{L}_r(x,y,\lambda, \dot{x}, \dot{y}, \dot{\lambda})$$

$$= \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) + mgx - \lambda f(x,y)$$

$$\boxed{\Rightarrow} \quad \frac{d}{dt} \left( \frac{\partial \mathcal{L}_r}{\partial \dot{\lambda}} \right) = \frac{\partial \mathcal{L}_r}{\partial \lambda} = -f(x,y)$$

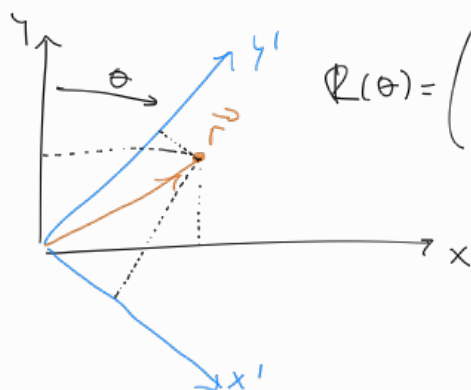


$$\begin{cases} x = l \cos \theta \\ y = l \sin \theta \end{cases} \rightarrow \text{La Variable Generalizada es } \theta$$

## Simetrías & Leyes de Conservación

Transformación Continua (grupo de Lie)

Rotaciones en el plano



$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$$\circ \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

Vamos la estructura de la Matriz de Rotaciones

$$R(0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Vea que  $R(\theta) = \cos\theta \text{ id}_2 + \sin\theta J$

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

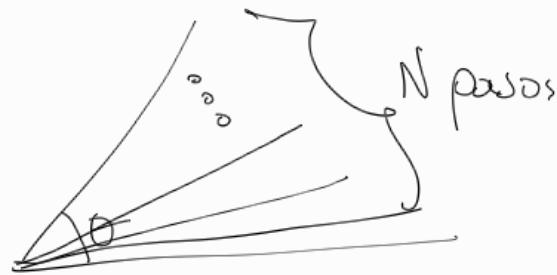
$$\Rightarrow J^2 = -\text{id}_2 \leadsto \text{Similar a } i^2 = -1$$

Entonces podemos tratarlo en el plano Complejo con Matrices

$$R(\theta) = \cos\theta \text{ id}_2 + \sin\theta J \longleftrightarrow \cos\theta + i \sin\theta = e^{i\theta}$$

i.e.  $\underbrace{\| R(\theta) = e^{\theta J} \|}_{\text{Generador Infinitesimal}}$

Generador Infinitesimal



$$\begin{pmatrix} \cos \Delta\theta & -\sin \Delta\theta \\ \sin \Delta\theta & \cos \Delta\theta \end{pmatrix} \approx \begin{pmatrix} 1 & -\Delta\theta \\ \Delta\theta & 1 \end{pmatrix}$$

$$= \text{id}_2 + \Delta\theta J$$

$$= \text{id}_2 + \frac{\theta}{N} J$$

i.e.  $\mathcal{J} \longrightarrow \text{'algebra de Lie'}$